



A systematic review and meta-analysis of electroencephalographic evidence and applied methods to cue-reactivity and cue exposure. Part II: Frequency analysis

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ABSTRACT

In this systematic review we report empirical evidence of electroencephalography (EEG) frequency analyses on the differences in processing addiction-related and neutral cues in individuals with addictive behaviors and control participants. A pre-registered (PROSPERO: CRD42024500485) PubMed and Web of Science search for cue reactivity and electroencephalography was conducted. Quality assessment was done with the Newcastle-Ottawa Scale. We identified eleven relevant studies. Methodological parameters across studies were presented. We could perform a meta-analysis for Δ -alpha power including three studies (233 individuals with addictive behaviors) comparing target and control cues. Meta-analyses reveal that in individuals with addictive behaviors target cues were not different compared to control cues ($k = 3$, Hedge's $g = -0.615 [-2.924, 1.693]$, $z = -0.523$, $p = .601$).

The limited number of studies and considerable variability in methodologies and parameters emphasize the necessity for additional research to validate these findings and elucidate the underlying mechanisms. This study highlights the critical importance of employing standardized methods in addiction research to improve the comparability and reliability of results.

1. Introduction

1.1. Cue reactivity and craving in addictive behaviors

Cue reactivity and craving are pivotal mechanisms involved in the development and maintenance of addictive behaviors including substance-related (e.g., use of alcohol or cocaine) and substance-unrelated (e.g., gambling or gaming, and candidate behavioral addictions as compulsive buying-shopping, problematic pornography use, and social-network use) disorders (Antons et al., 2020; Drummond, 2000). Cue reactivity refers to the phenomenon where exposure to drug-related cues elicits physiological and psychological responses in individuals with a history of addictive behavior (Brand et al., 2019; Loeber and Duka, 2009; Starcke et al., 2018). Craving, recognized as a

clinical feature of addictive disorders in the ICD-11, is a strong desire or urge to use a substance or engage in the specific non-substance-related behavior, which can be triggered by both internal states and external cues (Breiner et al., 1999; Carter and Tiffany, 1999). Both cue reactivity and craving are linked to engagement in these behaviors as well as relapse (Antons et al., 2019; Hawker et al., 2021; Serre et al., 2015; Vafaie and Kober, 2022).

Importantly, although each substance-related and behavioral addictions has its own specificities with respect to its phenomenology and neurobiological pathways (e.g., the direct effects of substances on the brain), current evidence and theoretical frameworks highlight core mechanisms, particularly with regard to cue reactivity, craving, and reinforcement learning processes (Brand et al., 2019, 2025), that are essential across both substance-related and behavioral addictions. These

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transdiagnostic perspectives suggest partially overlapping neural systems, especially within cortico-striatal circuits. However, it remains an open empirical question to what extent cortical oscillatory dynamics, as assessed by Electroencephalography (EEG) frequency analyses, are comparable across different types of addictive behaviors. The present review therefore follows an integrative approach while explicitly acknowledging potential heterogeneity between addiction domains.

Cue reactivity emerges through both associative and instrumental learning processes, where exposure to addiction-related cues activates conditioned responses (Carter and Tiffany, 1999; Hyman et al., 2006). It is assumed that a hypersensitization of the brain reward system is involved in the development of cue reactivity (Robinson and Berridge, 2001). These cues can be internal or external stimuli, such as visual, olfactory, auditory, or contextual elements previously linked to the behavior. Evidence suggests that even during early developmental stages, when individuals are only at risk of developing an addiction, certain forms of cue reactivity can already be observed (Diers et al., 2023). According to the Interaction of Person–Affect–Cognition–Execution (I-PACE) model and its specifications (Brand et al., 2019, 2025; Brandtner et al., 2021), awareness of addiction-related cues may, in the initial non-problematic stages, involve deliberate imaginal and verbal elaboration on the behavior (i.e., desire thinking). In later stages, however, these cues may provoke more intense and difficult-to-resist craving experiences. Over time, the responses tend to become increasingly automatic and habitual-like, with reduced conscious involvement in desire thinking. Conditioned stimuli not only evoke subjective craving but also measurable psychophysiological reactions, including variations in heart rate, changes in skin conductance, and altered neural activity in brain regions associated with reward, memory, and attentional processes (Antons et al., 2020; Starcke et al., 2018; Zilverstand et al., 2018). While these stage-related changes are well described on a theoretical level, it remains largely unclear how they are reflected in EEG frequency dynamics, including potential differences in alpha or beta band activity or their spatial distribution. Especially it should be considered that changes might be non-linear or asymmetrical. For example, cue-reactivity ratings have shown that individuals with at-risk show more positive image ratings use compared to individuals with non-problematic and pathological use (Antons et al., in press). This non-proportional development of liking (valence) and wanting (craving, arousal) might be mirrored in non-linear or asymmetrical changes in frequency bands (e.g., frontal alpha asymmetry when processing positive and negative stimuli (Perone and Vaughan, 2024)).

1.2. Neurobiological mechanisms of cue-reactivity

Studies utilizing functional magnetic resonance imaging (fMRI) have demonstrated that addictive-behavior-related cues activate brain regions as the mesolimbic reward pathway, including the nucleus accumbens and ventral tegmental area, as well as regions involved in emotional regulation and memory, such as the amygdala and hippocampus (Koob and Volkow, 2016). The ventral striatum as part of the mesolimbic reward pathway has been related with early stages of addictive behaviors, whereas the dorsal striatum is probably additionally involved in later stages (Antons et al., 2020; Starcke et al., 2018). These changes in neural process may represent the shift from reward-driven (“feels better”) behaviors to compulsive (“must do”) behaviors (Everitt and Robbins, 2005, 2013, 2016; Zhou et al., 2019). Besides neuroimaging, psychophysiological responses have been related to cue reactivity and can be potential psychophysiological marker of cue reactivity. EEG Power in specific frequency domains are such markers that have repeatedly investigated in cue-reactivity studies on addictions with EEG. However, the interpretation of EEG frequency changes across addictive behaviors requires caution. Different substances act through distinct pharmacological pathways and may differ in their acute, chronic, and withdrawal-related effects on cortical activity. For

example, stimulant use and nicotine use may both be associated with cue-reactivity effects, but they do not necessarily share identical baseline oscillatory profiles or neurophysiological mechanisms.

1.3. Approaches to studying the neurobiological mechanisms of cue reactivity

The primary methods used to examine neural mechanisms of cue reactivity are fMRI and EEG. fMRI offers detailed spatial information about brain regions involved in cue processing but has limited capacity to capture the fast temporal dynamics of neural activity. Frequency analyses of EEG, in contrast, provide insights into oscillatory processes across distinct frequency bands, which are closely linked to cognitive and affective functions. Although EEG is less sensitive to subcortical structures (e.g., ventral tegmental area, ventral striatum), frequency-based measures yield a valuable perspective on the rhythmic coordination of cortical activity during cue reactivity. As most previous reviews have concentrated on fMRI-based findings, the current review seeks to complement this work by synthesizing the growing literature on EEG frequency analyses of cue reactivity. Furthermore, while methodological guidelines for designing and reporting cue-reactivity paradigms are available for fMRI (Ekhtiari et al., 2022), comparable recommendations remain absent for frequency-analytic approaches.

1.4. Cue reactivity paradigms

Cue reactivity paradigms involve exposing individuals to both addiction-related and control cues. These paradigms vary significantly in their design, including the type and number of stimuli, the order and content of target and control stimuli, task timing and components, and cue display parameters. Typically, the dependent variables measured are subjective ratings of craving or urge, as well as psychophysiological and neural responses induced by different types of cues. The primary comparison of interest is often between the behavioral/neural responses to behavior-related cues versus neutral control cues. Additionally, comparisons between individuals with addictive behaviors and matched non- or casual-using control subjects, or between individuals with pathological, risky and non-problematic use, provide valuable insights. Researchers also correlate neural measures with subjective reports (e.g., craving ratings, addiction symptom severity) or objective performance metrics (e.g., reaction time, psychophysiological measures like skin conductance) (Chase et al., 2011; Jasinska et al., 2014; Yalachkov et al., 2012). However, there are considerable differences in the specific parameters used in cue-reactivity paradigms, especially those investigating event-related potentials (ERPs) and EEG Power in specific frequency domains. These differences include the number of trials, duration, size, and individualization of cues, all of which can influence the cue reactivity response. The choice of control cues is another critical factor. Control cues can range from appetitive cues (e.g., pleasant or erotic images to neutral cues (e.g., objects or pastoral scenes). While appetitive cues may yield more specific but less robust contrasts, neutral cues tend to produce greater effects with reduced specificity (Jasinska et al., 2014).

1.5. EEG Power in specific frequency domains

One key aspect of EEG analysis is the examination of power within specific frequency domains, which can reveal important information about brain function and states. EEG signals are typically divided into several frequency bands, each associated with different types of brain activity. These bands include delta (0.5–4 Hz), theta (4–8 Hz), alpha (8–13 Hz), beta (13–30 Hz), and gamma (30–100 Hz) frequencies. Sometimes alpha and beta bands are additionally divided in slow alpha (8–10 Hz) and fast alpha (11–13 Hz), as well as slow beta (13–18 Hz) and fast beta (18–30 Hz). A reduction of power in the EEG in a specific band following the onset of a stimulus is named event-related

desynchronization (ERD). Each frequency band is thought to reflect distinct cognitive processes and states of consciousness. For example, premotor cognitive processes such as response program updating was related to theta power while beta power was discussed as correlate of inhibitory control (Huster et al., 2013). Emotional, attention capturing stimuli induce higher levels of alpha desynchronization than neutral stimuli (De Cesare and Codispoti, 2011). In substance-use disorder research, an increase in beta power during cue exposure was associated with an increased cortical arousal or processing bias for drug-related cues (Knott et al., 2008; Littell et al., 2009). Self-reported craving was related to changes in beta, theta and delta power (Littell et al., 2009). Greater alpha ERD was reported to addiction specific stimuli compared to control stimuli (Cui et al., 2013). It should be noted that the functional interpretation of these frequency bands may vary depending on the specific addiction domain and task characteristics, which further limits direct comparability across studies.

1.6. Research question

To provide guidance for designing EEG cue reactivity studies, the aim of the current review and meta-analysis was to present an overview of the used cue reactivity task parameters, the applied EEG parameters, and to present meta-analytic evidence of EEG cue reactivity brain response across addictive behaviors. Given the limited number of available studies, we adopted a broad integrative perspective, while explicitly acknowledging that different addictive substances may differ in their pharmacological effects and baseline cortical modulations. Where possible, we additionally aimed to consider specific addictive behaviors separately. In the first part we have focused on event-related potentials (ERPs), and especially on the P300 and the late positive potential (LPP), while in this part we will focus on EEG Power in specific frequency domains (Diers et al., 2026).

2. MATERIAL AND METHODS

The systematic review and meta-analysis protocol was preregistered and published online (in January 2024) on the PROSPERO International prospective register of systematic reviews prior to data analysis (CRD42024500485; available from: https://www.crd.york.ac.uk/prosp/ero/display_record.php?ID=CRD42024500485).

Study characteristics were defined by specifying the PICOS (participants, interventions, comparators, outcomes, and study design) criteria presented in Table 1 to describe EEG paradigms used to investigate the population of substance or behavioral addictions during cue-reactivity (see supplemental table S1 for individual study details) (Smith et al., 2011). Additionally, all detailed processes such as identification and selection of the articles in the research process were carried out using the Preferred Reporting Items for Systematic Investigations and Meta-Analyses (PRISMA) guidelines (Liberati et al., 2009; Moher et al., 2009b, 2009a; Page et al., 2021) (see Fig. 1).

2.1. Search strategy

The search string was determined by consensus among MD and SA. The PubMed search was conducted in January 2024 updated in January 2025 and on March 20, 2026 to include the most recent literature with

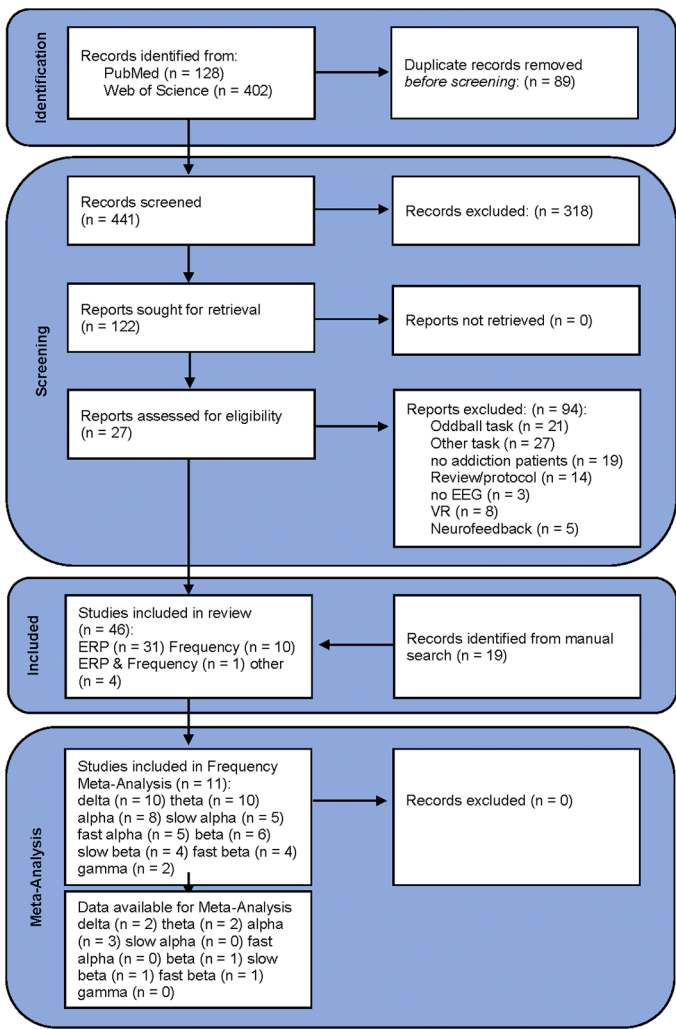


Fig. 1. Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) flow diagram.

the following search string: („cue-reactivity”[Title/Abstract] OR „cue exposure”[Title/Abstract]) AND (EEG[Title/Abstract] OR electroencephal* [Title/Abstract] OR electrocortical[Title/Abstract] OR "alpha band"[Title/Abstract] OR "beta band"[Title/Abstract] OR "gamma band"[Title/Abstract] OR "delta band"[Title/Abstract] OR "theta band"[Title/Abstract] OR "frequency band"[Title/Abstract] OR "frequency power"[Title/Abstract] OR "alpha power"[Title/Abstract] OR "beta power"[Title/Abstract] OR "gamma power"[Title/Abstract] OR "delta power"[Title/Abstract] OR "theta power"[Title/Abstract] OR "event related potential"[Title/Abstract] OR "evoked potential"[Title/Abstract]).

The Web of Science search was conducted with the following search string: (AB=(cue-reactivity OR cue exposure)) AND AB=(EEG OR electroencephal* OR electrocortical OR alpha band OR beta band OR gamma band OR delta band OR theta band OR frequency band OR frequency power OR alpha power OR beta power OR gamma power OR delta power OR theta power OR event related potential OR evoked potential).

Additionally, the references of retrieved and related articles were screened for further possible inclusions.

Two researchers performed independent initial searches. Search results from both data bases were combined and checked for duplicates based on DOI or PMID, if no DOI is available. Results with neither DOI, nor PMID were checked based on authors and title. After this, titles and abstracts were judged eligible independently from two raters (MD and

Table 1
PICOS criteria for the search.

Criteria	Definition
P	Population
I	Interventions
C	Control
O	Outcome
S	Study type

SA). The resolution of inconsistencies and full-text checks were performed in a consensus meeting involving both raters and the final list of included studies was also approved from both of them.

2.2. Inclusion criteria

We included studies that (A) were published in scholarly peer-reviewed journals up to March 20, 2026; (B) report original research (i.e. no reviews/ meta-analyses); (C) used EEG; (D) were conducted in humans; (E) used a paradigms designed to assess cue-reactivity responses; (F) included participants with tendencies towards a specific addiction domain; (G) had a cross-sectional design or report a baseline measurement in longitudinal studies.

2.3. Exclusion criteria

We excluded studies in which (A) no EEG measurement was reported; (B) non-humans were investigated; (C) a task interferes with cue reactivity (e.g. oddball task, mood or emotion induction, counting or button press for specific cues). Furthermore, we did not include studies that were (D) reported in a language other than English and (E) published only in the gray literature (e.g., conference papers or doctoral theses).

2.4. Data collection and analysis

We extracted general descriptors of the articles (name of first author, year of publication, journal name, title) and the investigated sample (age, sex, handedness, number of participants [each per group]).

For the task details of the cue-reactivity paradigm the following information were extracted from identified studies: total number of cues (target/control), number of blocks (combined target and control/target/control), trials per block, duration of events (fixation cross, cue, ITI, rating, other), order of blocks and trials, number of different cues (target/control), repeated presentation of each cue, repetition of paradigm, reason for repetition, and cue presentation details (background, size, location, kind of monitor, kind of computer, viewing distance, visual angle, pixel, dpi, software used for cue presentation), as well as instructions of paradigm and number of test trials.

For the cue details we were interested in the content of target cues and control cues, source of target cues and control cues, matching of physical features between cue categories, and validation of cues (target/control).

For the EEG recording and analyzing parameters we extracted the company of EEG system, the acquisition and analyses software, number of electrodes, kind of electrodes, ground electrode, reference electrode during measurement and used for analysis, applied impedance, digitization parameters, filters used during recording and preprocessing, and number of included/excluded trials.

For the meta-analysis the necessary data were either extracted from the original articles or provided by the articles' authors. The extracted data were the power for delta, theta, alpha and beta and gamma bands (mean, standard deviation) for target and control cues separately for patients and controls, as well as the sample/group size. To enhance comparability across studies, we prioritized the extraction of Δ -power measures (i.e., differences between target and control conditions). In cases where other metrics (e.g., ERD) were reported, these were transformed into Δ -power where sufficient information was available. However, we acknowledge that such transformations do not fully harmonize mathematically distinct measures and may introduce additional variability. For studies employing complex or composite stimuli (e.g., no smoking signs combining reward-related and inhibitory components), data extraction focused on conditions isolating reward-related processing, such as smoking symbols (SSs) contrasted with matched control activity symbols (CSs), in order to ensure comparability with classical cue-reactivity paradigms. Data extraction was performed by

one researcher (MD). In case an eligible article did not report necessary information to calculate the relevant effect sizes, we contacted the authors via email to request the information/data. In case of no response authors were reminded after 3 weeks, and after additional 3 weeks, this data point was marked as missing information. Excel spreadsheets were used to record study selection and extracted data. Data were analyzed using the statistics software Jamovi (version 2.3.28.0) ([The jamovi project \(2022\), n.d.](#)) based on R (version 4.1) ([R Core Team 2021, n.d.](#)). For meta-analysis, we used the MAJOR package based on the R package 'metafor' ([Viechtbauer, W., n.d.](#)). We used restricted Maximum-Likelihood estimator based on the mean scores (M) and standard deviations (SD) to calculate effect sizes comparing delta, theta, alpha or beta power for target and neutral control cues with each other and each between patient and control groups. Statistical significance was defined as $p < .05$, two-tailed. To provide a more generalizable model estimate, we used random-effects models in all cases. Heterogeneity was quantified using τ^2 and Q-tests. Forest plots were used for data visualization. Cook's distances were used to identify outliers and/or studies that were highly influential to models. Studies with a Cook's distance larger than the median plus six times the IQR of the Cook's distances were considered highly influential.

2.5. Risk of bias (quality) assessment

Risk of bias was assessed using the Newcastle-Ottawa Scale (NOS) for quality assessment in case-control studies ([Wells et al., 2000](#)). All eligible studies were assessed against the criteria 'selection' (max. 4 points), 'comparability' (max. 2 points), and 'exposure' (max. 3 points), resulting in a quality score between 0 and 9 (see [supplemental material table S2](#)).

3. Results

3.1. Study selection

Overall, 27 relevant articles were identified with the screening procedure and 19 additional articles were included from the reference lists of the included articles. From these 46 articles, 31 papers used ERP analyses, ten frequency analyses, one study used ERP and frequency analyses, and four used other analyses. The ERP data was presented elsewhere ([Diers et al., 2026](#)). Therefore, eleven papers were reviewed for the EEG Power in specific frequency domains part presented in this article. For the Meta-Analysis we had data for the Δ -alpha power from one study (transformed from alpha EDR), two studies reported Δ -alpha, Δ -beta, Δ -delta and Δ -theta power, although one of these study divided beta in slow and fast beta. One additional study reported only relative beta and theta power but no values to transform these to Δ -power. We therefore performed analyses only for the three studies reporting Δ -alpha power or provided the values to transform to Δ -alpha power. All three of these studies reported Δ -alpha power for target and control stimuli, only for patients.

3.2. Cues: presentation and display parameters, content, matching and validation

The number of used cues (target and controls) was only one per condition in 64% ($n = 7$) of the studies ([Table 2](#)). Six studies (55%) used a randomized presentation of target and control cues, eight studies (73%) used different blocks for target and control cues. Cue presentation time was quite variable between studies (range 2 ms-30 min). Seven studies included a rating within the paradigm. Most studies ($n = 10$) showed each cue only once. The cue display parameters and presentation software were basically not described (see [table S3](#)). The instructions were reported from 6 of the 11 studies ([Tables 3 and 4](#)). All studies used addictive-behavior-related cues as target and neutral cues as control stimuli ([table 4](#)). Other control categories were frequently

Table 2
Cue presentation parameters.

	n of cues t/c	n of blocks t&c/t/c	trials/block n	fixation (s)	cue (min)	ITI (ms)	rating	other	order of blocks	order of trials	number of different cues t/c	a	b
(Kim et al., 2023) [1]	5/5	1/0/0	5	10 s	20 s	0	%	t & c in one trial	random	fixed	5/5	1	0
(Bauer and Krantzler, 1994) [2]	1/1 + 1	0/1/1 + 1	1	%	5	%	every 30 s	%	pseudo-random	%	1/1 + 1	1	0
(Herish et al., 1995) [3]	1/1 + 1	0/1/1 + 1	1	%	5	%	every 30 s	%	pseudo-random	%	1/1 + 1	1	0
(Liu et al., 1998) [4]	1/1 + 1 + 1	0/1/1 + 1 + 1	1	%	30/30 + 5 + 7	%	every 10 min	%	random	%	1/1	1	0
(Reid et al., 2003) [5]	1/1 + 1 + 1	0/1/1 + 1 + 1	1	%	15–20/15–20 + 5 + 5	%	before + after blocks	%	fixed	%	1/1	1	0
(Lu et al., 2021) [6]	1/1 + 1 + 1	0/1/1 + 1 + 1	1	%	2.5/2.5 + 3 + 3	%	after exp.	%	fixed	%	1/1	1	0
(Cui et al., 2013) [7]	48/48 + 48 + 48	1/0/0	96	%	4 s	3–5 s	%	%	%	random	24/24 + 24 + 24	2	0
(V. Knott et al., 2008) [8]	1 + 1 + 1/1 + 1 + 1	0/3/3	1	%	1 + 1	%	%	%	counter-balanced	%	1 + 1 + 1/1 + 1 + 1	1	0
(Kroczeck et al., 2024) [9]	1/1 + 1 + 1	0/1/1 + 1 + 1	1	%	5	%	during exp.: 10 ratings	%	fixed	%	1/1	1	0
(Littel et al., 2009) [10]	1/1	0/1/1	1	%	30 s	%	before + after exp.	%	fixed	%	1/1	1	0
(Li et al., 2022) [11]	108 + 108/108 + 108 + 48	4/0/0	120	1–5	2 s	%	%	%	balanced	pseudo-random	27 + 27/27 + 27 + 12	4	0

Note: n = number; t = target; c = control; ms = milliseconds; ITI = inter trial interval; (a) presentation of each cue; (b) repetition of paradigm; e.g. [1] is short number of publication for other tables and figures.

Table 3
Used instructions for cue-reactivity paradigms and used test trials.

	instructions
[1]	%
[2]	%
[3]	They were told that they would view three different 5-min films on a television screen and were asked to try to keep still, focus their eyes on the screen, and minimize eye blinking.
[4]	Subjects were instructed to keep their eyes focused on the stimulus complex and to refrain from unnecessary talking and movement (including eye blinks) except when providing subjective self-reports. To increase the likelihood of eliciting craving in the cocaine abuser group, subjects in that group were told that they would be allowed to self-administer (by insufflation) the cocaine in view after completion of the experimental procedures (including acquisition of brain scans).
[5]	the patient was instructed to pick-up and view each of the cocaine-related items over a period of approximately 5 min, and then a 5-minute video consisting of scenes specific to the purchase and smoking of crack cocaine was presented. During the 5 min guided imagery session, the patient was instructed to close his eyes and recall pleasurable memories associated with cocaine, to think about the items they had just viewed, and to "imagine the taste of the cocaine, the desire to use becoming stronger, and to know that there is no good reason not to use at this time."
[6]	%
[7]	%
[8]	Participants were instructed to close their eyes and listen carefully to each of the 60-s scripts and to imagine themselves in the scene both during the script and in the 60-s active imagery period that followed each script.
[9]	Item exposure task began with the opening of box, including instructions for visual, tactile, and olfactory inspection of the items. The task ended after the instruction to use the items as usual.
[10]	Instructions were to sit relaxed and still and to carefully attend to the cues without employing distracting thoughts.
[11]	%

eyes-closed and eyes-open baselines but also erotic pictures or signs. Matching and validation of stimuli was only reported in two studies.

3.3. EEG parameters

The used EEG system, recording and analyzing software were quite variable and not always reported (Table 5). The number of used electrodes normally was between 19 and 64 electrodes in six studies (range 5–129), type Ag/AgCl or ActiCAP. The used ground electrode was either AFz or probably on a similar location (a mid fronto-central electrode) or not reported (n = 7). During recording the reference electrode was quite variable between linked ear lobes or FCz and five times not reported. During analysis reference was kept or referenced to linked ear lobes / mastoids (five times) or average reference (four times). During measurement Impedance was kept in five studies ≤ 5 kΩ, in three studies ≤ 10 kΩ (three times not reported). Filter settings during recording and analysis were also very variable. Six studies (55%) did not report the number of included epochs.

3.4. Borders of frequency bands and reporting of results

The boundaries of the various frequency bands are largely comparable across studies (see Table 6). The main difference is that some studies divide the alpha and beta bands into slow and fast sub-bands, and not all studies provide values for the entire band. The most divergent aspect is the reported measures. These are expressed in absolute power, relative power, total power, Δ-power, alpha EDR, power or band power (not further described), as well as averaged log-transformed amplitude. Absolute and relative power are sometimes log-transformed for normalization, although the base of the logarithmic transformation is not always specified. This makes comparing results challenging, and a transformation without additional information from the authors is often not feasible. Importantly, these measures are mathematically distinct and reflect different aspects of neural activity (e.g., baseline-referenced vs. relative or normalized indices), and are

Table 4

Cue content, matching, and validation.

	modality	t content	c content 1	c content 2	c content 3	c content 4	t source	c source	Matching Physical Features between Cue Categories	Cue validation (t/c)
[1]	t: audio-video, c: images	alcohol, individual choice of either beer, wine, whiskey, soju, or makgeolli	non-alcoholic	%	%	%	%	%	nonalcoholic cues matching color of craving-inducing stimuli	focus group of 10 alcoholic patients
[2]	video	cocaine, use, pre-paration, administration	neutral, pastoral scenes	erotic, woman undressing dancing around car	%	%	%	%	%	%
[3]	video	cocaine, use, pre-paration, administration	neutral, pastoral scenes	erotic, woman undressing dancing around car	%	%	%	%	%	%
[4]	video	cocaine use	neutral objects	eyes-closed baseline	eyes open baseline	%	Research Technology ranch, National Institute on Drug Abuse (NIDA)	%	%	%
[5]	object handling, video, guided imagery	crack pipes, screens, lighters, \$20-dollar bill, fake crack rocks, preparation, smoking	shells, rocks, pine cone	eyes-closed baseline	eyes open baseline	%	%	%	%	%
[6]	object viewing	drug tools	five grocery items	eyes-closed baseline	eyes open baseline	%	%	%	%	%
[7]	images	smoking	neutral	pleasant	unpleasant	%	(Carter et al., 2006; Gilbert et al., 2005) (Tiffany & Drobos, 1990)	(Lang et al., 2005)	%	%
[8]	auditory scripts	% desire to smoke: urge vs no-urge	neutral (urge vs no-urge)	pleasant (urge vs no-urge)	unpleasant (urge vs no-urge)	%	%	%	%	%
[9]	object handling	cigarettes, lighter, ashtray	(pens, sharpener, rubber, white paper)	eyes open pre task baseline	eyes open post task baseline	%	%	%	%	%
[10]	object viewing	lit cigarette on the same dish	pen on a small dish	%	%	%	%	%	%	%
[11]	signs	smoking symbols (SSs)	control activity signs (CSs)	no smoking signs (NSSs)	control no-activity signs (CNSs)	animal images	downloaded 54 paired no smoking signs (NSSs) and smoking symbols (SSs), 54 paired control no-activity signs (CNSs) and control activity signs (CSs) from open source on the internet	14 animal images from open source on the internet	Selection criteria: 1) CS and animal images do not induce any desire for smoking; 2) getting rid of 10 NSS and 10 CNS showing the highest SD; 3) pairing CNS and CS based on the content style (e.g., color, lightness, complexity, etc); 4) selected 27 NSS/ CNS showing the lowest difference of pleasure and prohibition; 5) getting rid of two animal images to make consistent pleasure level to the types of CNS/ CS images	10 smokers rated 1) valence; 2) desire for cigarette smoking; 3) to what extent the image stops them from doing something, (effect of prohibition) (scale of 1–9).

Note: t = target; c = control; IAPS = International Affective Picture System.

Table 5
EEG assessment and preprocessing parameters.

	EEG Company	Recording (R) and analyzing (A) software	No. Elec- trodes	Kind of electrodes	Ground	Reference recording	Reference analysis	Impe- dance	digitized	Filter recording	Filter	No. included trials
[1]	PolyG-A, (Laxtha, Daejeon, Republic of Korea)	A: NeuroGuide software (Applied Neuroscience; St. Petersburg, Florida, U.S.)	8		%	%	average reference	≤ 10 kΩ	256 Hz	band pass 1–600 Hz	%	30 artifact-free epochs (epoch length: 2048 s) were determined for each participant at least 50 epochs
[2]	Grass Instrument Co. Model 7PI22 amplifiers (Microstar Laboratories Inc.) 2400/6 A-D converter	%	5	Ag/AgCl electrodes	mid- forehead	%	%	≤ 5 kΩ	100 Hz	band pass 0.01–35 Hz	%	
[3]	Grass Instrument Co. Model 7PI22 amplifiers (IZbit Microstar Laboratories Inc.) 2400/6 A-D converter	A: FORTRAN program	5	Ag/AgCl electrodes	mid- forehead	linked ear lobes	%	≤ 5 kΩ	100 Hz	band pass 0.01–35 Hz	%	
[4]	Grass Model 12 Neurodata Acquisition System	%	22	Ag/AgCl electrodes	%	tip of nose	linked ear lobes	≤ 5 kΩ	256 Hz	band pass 0.1–50 Hz	%	maximum of 60 artifact-free epochs (2 min), minimum number of artifact-free epochs 40 (1 min and 20 s)
[5]	Lexicor Biograph System 2.0 (Boulder, CO, U.S.)	%	19	silver chloride disc electrodes	%	linked ear lobes	%	≤ 5 kΩ	128 Hz	band pass 0.016–100 Hz, notch 60 Hz	%	
[6]	(Brain Products GmbH, Germany)	A: MATLAB 2017b and EEGLAB toolbox	64	Ag/AgCl electrodes	AFz	FCz	average reference	≤ 10 kΩ	1000 Hz	%	high pass 0.5 Hz, low pass 100 Hz, notch 50 Hz	
[7]	Geodesic EEG System 200 (Electrical Geodesics Inc.)	A: BESA v 5.1.8.10, (MEGIS Software GmbH), BrainVision Analyzer software v 1.05.0005, (Brain Products GmbH)	129	geodesic sensor net	%	%	average reference	%	250 Hz	band pass 0.1–100 Hz	%	If < 90% of channels in an epoch were artifact free, epoch was excluded from analyses. → loss of approximately 17% of all epochs
[8]	%	%	%	%	%	linked ear lobes	%	%	256 Hz	band pass 0.1–30 Hz notch 50 Hz	%	
[9]	BrainAmp DC amplifier, (Brain Products GmbH, Gilching, Germany)	R: BrainVision Recorder, (Brain Products GmbH, Gilching, Germany) A: BrainVision Analyzer, V 2.2.0, (Brain Products GmbH, Gilching, Germany); EEGLAB (Delorme & Makeig, 2004), (Matlab, The MathWorks Inc., Natick, USA) toolbox	22	passive EEG electrodes (Brain Products GmbH, Gilching, Germany)	%	FCz	average reference	≤ 10 kΩ	1000 Hz	band pass 0.1–30 Hz notch 50 Hz	band pass? -40 Hz	
[10]	digital BioSemi Active-Two system	%	38	active Ag/AgCl electrodes	%	%	linked mastoids	%	1024 Hz	low pass 134 Hz	band pass 0.1–40 Hz (24 dB/octave roll off)	mean number artifact-free segments: neutral condition: smokers 28.41, ex-smokers 29.29, smoking condition: smokers 29.18, ex-smokers 29.10 %
[11]	SynAmps 2 amplifier (NeuroScan, Charlotte, NC, USA)	A: Scan 4.0. (Neuroscan Inc.), Brain Electrical Source Analysis Statistic 1.0	64	Ag/AgCl electrodes	AFz and nose	%	%	≤ 5 kΩ	500 Hz, 24-bit resolution	Alternating current signals (0.05–100 Hz) continually recorded	band pass 0.5–25 Hz	

Hz = hertz, dB = decibel.

Table 6

Frequencies of specific bands and measure of reported results.

	Frequenz delta	in Hz theta	alpha total	slow	fast	beta total	slow	fast	gamma	measure
[1]	1–4	4–7	8–15	%	%	16–31	%	%	32–59	absolute and relative power
[2]	0.78–3.8	4.6–7.0	%	7.8–10.1	10.9–12.5	%	13.2–17.9	18.6–27.6	%	absolute and relative power, natural log transformed
[3]	0.7–4.0	4.7–7.0	%	7.8–10.1	10.9–12.5	%	13.2–17.9	18.6–27.6	%	probably absolute power
[4]	1–3.5	4–7.5	%	8–10.5	11–13	%	13.5–17	17.5–30	30.5–50	absolute power transformed into a percent change score → probably Δ -power
[5]	1.5–3.5	3.5–7.5	7.5–12.5	%	%	12.5–25	%	%	%	absolute and relative power, total power mean frequency, inter and intra-hemispheric coherence and symmetry
[6]	0.1–4	4–8	8–13	%	%	13–25	%	%	%	absolute power, Δ -power, mean frequency
[7]	%	%	8–12	%	%	%	%	%	%	alpha EDR
[8]	0.75–3.75	3.75–7.75	7.75–13.75	7.75–10.75	10.75–13.75	13.75–29.75	%	%	%	average log transformed amplitude in six frequency bands
[9]	0.5–4	4–8	8–12	%	%	13–30	%	%	%	power, log-transformed
[10]	0.75–3.75	3.75–7.75	7.75–13.75	7.75–10.75	10.75–13.75	13.75–29.75	%	%	%	power, log-transformed
[11]	1–3.9	4–7.9	8–12	%	%	%	12.1–20	20.1–30	%	band power

Hz = hertz; EDR = event-related desynchronization.

therefore not directly interchangeable. This heterogeneity in signal processing approaches limits direct comparability across studies and complicates the integration of results in meta-analytic frameworks, even when transformation into common metrics such as Δ -power is attempted, as such transformations cannot fully resolve differences in signal definition and scaling. In addition, heterogeneity also arises from differences in stimulus design. In studies employing composite stimuli (e.g., no smoking signs combining reward-related and inhibitory components), analyses in the present meta-analysis were based on stimulus components isolating reward-related processing (e.g., smoking symbols contrasted with control activity symbols), rather than on the combined stimuli.

3.5. Frequency changes during cue exposure

3.5.1. Alpha band

In response to addictive behavior-specific cues compared to control cues greater alpha ERD was reported in smokers (Cui et al., 2013). Similar, greater alpha power reductions for addiction-specific cues compared to rest or control cues were reported in smokers (Krocze et al., 2024) and individuals with cocaine use disorder (Bauer and Kranzler, 1994; Hersh et al., 1995; Liu et al., 1998).

3.5.2. Beta band

Compared to control participants, individuals with addictive behaviors showed an increased beta power for target cues compared to control cues in smokers (Krocze et al., 2024; Littell et al., 2009) and individuals with alcohol use disorder (Kim et al., 2023). Addictive behavior-specific paraphernalia handling, video viewing, or guided imagery compared to neutral paraphernalia handling, video viewing, or guided imagery lead to increased beta power in individuals with cocaine use disorder too (Reid et al., 2003).

3.5.3. Delta band

Delta power increased from rest to target cues in smokers (Krocze et al., 2024). During guided imagery of addiction-specific paraphernalia handling compared to neutral paraphernalia handling increased delta power in individuals with cocaine use disorder was reported too (Reid et al., 2003).

3.5.4. Theta band

The theta power was higher for target cues compared to rest in smokers (Krocze et al., 2024).

3.5.5. Contrary findings in beta, delta and theta bands

Contrary, in individuals with cocaine use disorder a greater absolute power during neutral compared to cocaine and erotic cues was reported in delta, theta, slow beta, and fast beta frequency bands (Bauer and Kranzler, 1994; Hersh et al., 1995). Individuals with cocaine use disorder and control participants showed similar responses (Bauer and Kranzler, 1994).

3.6. Meta-analytic results

One meta-analysis was conducted for Δ -alpha power: (1) comparisons between target and control cues for patients (including 3 effect sizes). Table 7 shows the investigated addiction group and the number of subjects per group.

For patients the results (see Fig. 2) indicate that responses towards target cues were not different from the control cues ($k = 3$, Hedge's $g = -0.615$ [-2.924, 1.693], $z = -0.523$, $p = .601$). According to the Q-test, the true outcomes appear to be heterogeneous ($Q(2) = 173.062$, $p < .0001$, $\tau^2 = 4.096$, $I^2 = 98.687\%$). Hence, although the average outcome is estimated to be negative, in some studies the true outcome may in fact be positive. An examination of the studentized residuals revealed that one study (Cui et al., 2013) had a value larger than ± 2.394 and may be a potential outlier in the context of this model. According to the Cook's distances, none of the studies could be considered to be overly influential. The regression test indicated funnel plot asymmetry ($p < 0.0001$) but not the rank correlation test ($p = 0.333$). However, the meta-analytic estimate should be interpreted with caution, as it is based on a small number of studies ($k = 3$) and is associated with very high heterogeneity ($I^2 = 98.687\%$), which may indicate substantial between-study variability and limited estimation stability.

4. Discussion

This systematic review identified 46 relevant papers, of which 11 reported frequency analysis. The meta-analysis of three studies revealed no difference in Δ -alpha power in individuals with addictive behaviors when processing addiction-related cues compared to control cues, with a small effect size. This result is not consistent with other studies descriptively included in the current review, showing that cue reactivity is associated with a decrease of alpha power (Bauer and Kranzler, 1994; Cui et al., 2013; Hersh et al., 1995; Krocze et al., 2024; Liu et al., 1998) and an increase of beta power (Kim et al., 2023; Krocze et al., 2024; Littell et al., 2009; Reid et al., 2003). Given the small number of studies

Table 7
number of subjects per group and investigated addiction for the studies included in the meta-analysis.

author year	kind of addiction	number addicts/ controls	age m±sd addicts / controls	sex m/f / addicts/ controls	handedness r/l / addicts/ controls	(cluster of) electrode(s) used for meta-analysis
[6] (Lu et al., 2021)	methamphetamine	21/0	33.71 ± 3.96 / %	21/0 / %	21/0 / %	Pz/3/4/5/6/7/8, CP1/2/3/4/5/6
[7] (Cui et al., 2013)	smokers	179/0	45.2 ± 10.5 / %	117/62 / %	%	P1/2/3/4/5/6/7/8, PO2/3/4/5/6/7, O1/2/3/4
[11] (Li et al., 2022)	smokers	33/0	23.76 2.88 / %	33/0 / %	%	AF3/, Fz/1/2/3/4/5/6/7/8, FT7/8, FCz/1/2/3/4/5/6/, T7/8, Cz/1/2/3/4/5/6, TP7/8, CPz/1/2/3/4/5/6, Pz/1/2/3/4/5/6/7/8, POz/3/4/5/6/7/8, Oz/1/2

Note: m = mean, sd = standard deviation, m = male, f = female, r = right, l = left

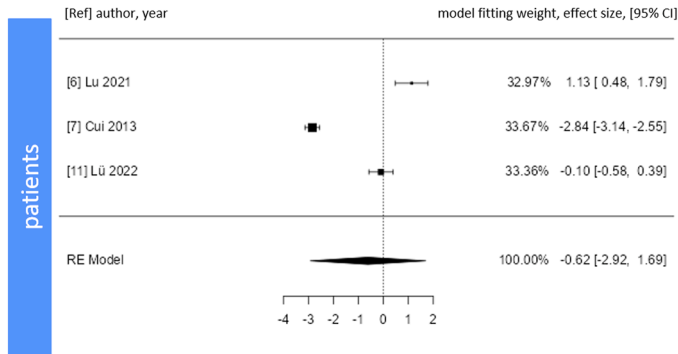


Fig. 2. Forest plots Δ -alpha power for patients comparing target and control cues. Effect sizes are Hedge's g. Negative effect sizes indicate that target cues showed lower Δ -alpha power compared to control cues.

included in the meta-analysis and the high heterogeneity, these findings should be interpreted with caution.

4.1. Alpha and beta power as marker for increased arousal and motivational significance

In general, alpha waves are observed during relaxed wakefulness and a decrease in alpha power is associated with cognitive tasks, arousal and an increase in beta power. In cue-reactivity studies, the decrease in alpha power for addictive behavior-specific cues compared to neutral control cues was similar to the decrease for pleasant cues compared to control/neutral cues (Bauer and Kranzler, 1994; Cui et al., 2013; Hersh et al., 1995). This finding is relevant as it provides valuable insight into the brain mechanisms activated in individuals with addiction when exposed to addictive behavior-specific cues. Consistent with current models of addiction, addiction-specific cues have high levels of motivational significance as other arousing pleasant (and unpleasant) stimuli. The high motivational significance is also evident in LPP responses for these cues (Cui et al., 2013).

Overall, beta power is associated with increased arousal (Niedermeyer and Silva, 2005), as well as key functions such as perception, cognition, orienting response, attentive behavior, and increased activation within an attentional alertness/vigilance network (Egner and Gruzelier, 2001, 2004; Haenschel et al., 2000; Singer, 1993; Wróbel, 2000). Notably, increased beta power during the processing of addiction-specific cues compared to control cues has been linked to factors such as inhibitory control (Krocze et al., 2024), impulsivity, and attention biases, which may contribute to craving when exposed to addictive stimuli (Fecteau et al., 2010). Beyond addictive behavior-specific cue exposure, beta power increases have also been observed in the context of cocaine craving-related guided imagery (Reid et al., 2003), exposure to smoking urge scripts (Knott et al., 2008), and the administration of nicotine (Domino, 2003; Teneggi et al., 2004) and cocaine (Herning et al., 1985, 1994; Reid et al., 2006).

In the context of cue reactivity, enhanced beta power may indicate a processing bias toward addictive behavior-related stimuli. Processing bias refers to the heightened processing of addictive behavior-specific stimuli, which results from their strong acquisition of incentive motivational properties (Robinson and Berridge, 2003). Through the repetitive execution of addictive behaviors, addiction-specific stimuli gain salience and occupy larger amounts of attentional resources compared to other (rewarding) stimuli (Franken, 2003; Robinson and Berridge, 2003). For individuals with addictive behaviors, the processing bias in cue reactivity is evident in the later components of ERP, such as the P300 and the LPP (see part I of this review). It is plausible that both arousal and (motivated) attention modulate both beta activity and LPP. Indeed, a direct association of beta power and P300 amplitudes could be shown (Egner and Gruzelier, 2001, 2004). In cue reactivity, the increase in beta power likely indicate an enhanced processing of addictive behavior-specific cues, highlighting their processing bias.

The contrary findings (Bauer and Kranzler, 1994; Hersh et al., 1995) reported above might be due to the relatively excellent health of that sample and the exclusion of subjects with confounding or comorbid conditions, including psychosis, depression, head injury, seizures, medication usage, and other drug/alcohol dependence. Another possibility is that any potent appetitive stimulus, whether associated with cocaine or not, evoked an increase in desire for cocaine and a decrease in EEG power.

4.2. Cue presentation in frequency band studies

In contrast to classical cue-reactivity paradigms, studies assessing frequency bands often employ a single stimulus presented over an extended period (e.g., 30 min), whereas traditional paradigms typically use multiple stimuli presented briefly (a few seconds each). These methodological differences may influence neural processing, including sustained versus transient attentional engagement, and could affect outcomes such as cognition, craving, and the magnitude of the cue-reactivity response. Future research should systematically examine how variations in stimulus duration and presentation format shape these mechanisms to refine the interpretation of cue-reactivity findings.

In this context, the dynamics of cue reactivity and craving must be considered, as their temporal patterns significantly influence cognitive and behavioral outcomes. Research indicates that cue-induced craving exhibits varying neural activations over time, with regions such as the amygdala and anterior cingulate cortex showing differential engagement during the craving response. Understanding these temporal dynamics is crucial for developing effective interventions targeting craving and relapse prevention (Jafakesh et al., 2022).

4.3. Limitations

While these findings provide a framework for understanding frequency-based cue reactivity, several conceptual and methodological limitations need to be considered. A critical conceptual issue concerns the aggregation of different addiction types. While transdiagnostic

models propose shared mechanisms underlying cue reactivity across substance-related and behavioral addictions, it is currently unclear whether these similarities extend to cortical oscillatory dynamics. Given the limited number of available studies and the heterogeneity of paradigms, the present results should therefore be interpreted with caution. Future research should aim to directly compare EEG frequency responses across specific addiction domains to determine the degree of neurophysiological overlap versus specificity.

An additional limitation concerns the aggregation of studies across different substance classes. Substances such as cocaine and nicotine differ substantially in their mechanisms of action and may also exert different baseline effects on cortical oscillations. Accordingly, combining these studies does not imply neurophysiological equivalence across substance classes but similarities in the mechanism of cue reactivity. In the present review, synthesis was primarily guided by the shared cue-reactivity paradigm and by the within-study contrast between addiction-related and control cues rather than by absolute EEG power levels across studies. Nevertheless, substance-specific baseline modulations, withdrawal state, chronicity of use, and potential neurotoxic effects likely contribute to the heterogeneity of findings and should be addressed more systematically in future work.

Another important limitation relates to the lack of differentiation between stages of addictive behavior. Theoretical models such as the I-PACE framework propose a transition from more goal-directed to more habitual and compulsive patterns of behavior, which may be accompanied by changes in neural processing. However, the studies included in the present review did not systematically assess or report clinical stages, severity levels, or progression markers that would allow for a stage-specific analysis of EEG frequency responses. As a result, it remains unclear whether such transitions are associated with distinct and potentially non-linear alterations in alpha or beta band activity or their topography. Future studies should explicitly investigate stage-dependent differences in oscillatory activity, ideally combining EEG measures with well-defined clinical or dimensional markers of addiction progression.

A further methodological limitation concerns the heterogeneity of EEG signal processing metrics across studies. Measures such as absolute power, relative power, and ERD are mathematically distinct and reflect different aspects of neural activity. Although we attempted to harmonize results by focusing on Δ -power (i.e., within-study contrasts between target and control conditions), including transformation of ERD measures where possible, these approaches do not fully resolve underlying differences in signal definition and scaling. Consequently, the meta-analytic integration of such measures should be interpreted with caution, as these methodological differences likely contribute to the observed heterogeneity.

Another important source of heterogeneity concerns the nature of the stimuli used in cue-reactivity paradigms. While most studies employed addiction-related cues designed to elicit craving through incentive sensitization, some studies used more complex stimuli combining reward-related and inhibitory components. In the present meta-analysis, we addressed this by focusing on stimulus components isolating reward-related processing (e.g., smoking symbols vs. control symbols) rather than pooling composite stimuli directly. Nevertheless, residual differences in task design and cognitive demands (e.g., involvement of inhibitory control processes) may contribute to variability across studies.

A major limitation of the present meta-analysis is the small number of included studies combined with very high heterogeneity. With only three studies contributing to the meta-analysis and an I^2 value close to 100%, the estimation of the pooled effect size is likely to be unstable and sensitive to individual study characteristics. Although a random-effects model was applied, such conditions substantially limit the robustness and generalizability of the findings. Therefore, the meta-analytic results should be considered exploratory and interpreted with caution until a larger body of comparable studies becomes available.

4.4. Further methodological considerations

A major issue in the present meta-analyses was the heterogeneous presentation of results. Until a clear consensus is established regarding which values should be used for reporting, it would be beneficial to provide a table containing all necessary data to allow transformation of results into different formats. This would facilitate comparisons across studies. For the current analyses, we chose to use Δ -power values. In addition, future studies should report substance-related characteristics more systematically, including current use status, abstinence or withdrawal duration, and other factors that may influence baseline cortical activity, in order to improve comparability across substance classes.

The limited number of studies included in these meta-analyses does not allow for strong recommendations regarding design parameters. However, for frequency analyses, employing a cluster of electrodes may be preferable to relying on a single electrode. Additionally, presenting multiple target and control cues could help capture attention and might enhance brain responses.

Intraindividual differences including age and memory performance need to be considered when assessing frequency bands (Klimesch, 1999).

5. Conclusions

The results of this study show that addiction-specific cues might lead to a decrease of alpha power similar to pleasant cues. However, as the numbers of included studies is low further research is needed and should be extended to the other frequency bands, especially to the beta band. Another focus should be separate analyses for specific addictions types and the specification of the brain localization of the effects.

Authors' contribution

MD and SA: study concept and design; MD: analysis and interpretation of data; MD: statistical analysis; MD: writing - original draft; MD, MB and SA: writing - review and editing. All authors had full access to all data in the study and take responsibility for the integrity of the data and the accuracy of the data analysis. All authors discussed the results, commented on the manuscript and approved the final version of the manuscript.

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Declaration of Competing Interest

The authors declare no conflict of interest.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.neubiorev.2026.106767](https://doi.org/10.1016/j.neubiorev.2026.106767).

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